C1

Case-mix-adjusted length of stay in FONASA GRD hospitals, 2023-2024

Business Intelligence IIB423T-1 · GROUP_01 · Matías López, Cristóbal Ramírez · 29 September 2026

1,719,702 inpatient stays · 72 hospitals · FONASA public GRD files

Decision and question

User and decision

FONASA's GRD financing team and hospital bed managers decide which hospitals and GRD groups to prioritise for a review of prolonged stays and discharge processes.

Measured now vs desired

Desired: fewer bed-days beyond what case mix explains, without worse outcomes (not observable here). Measured: IEMA with interval, % of bed-days in extreme stays, outliers observed/expected.

Main question

Which GRD hospitals show a stay systematically different from the network for their case mix, and how much bed use is in extreme stays? (2023-2024)

- How robust is the benchmark to its choices?
- Which GRDs and episodes concentrate extreme-stay bed-days?
- Is the hospital signal stable between 2023 and 2024?
- Which admission-time fields vary with extreme stays?

Data, preparation and quality

One row = one GRD-coded discharge, not a patient or a bed-day

2,125,400

episodes in the two FONASA extracts (unchanged)

1,719,702

inpatient stays analysed (CMA and 113 invalid excluded)

72 / 72

hospital codes found in the MINSAL facilities file (71 in FONASA's); m:1 join, rows unchanged

0.14%

identical rows, retained: no episode ID to prove duplication

Clean	No negative stays; same-day stays (LOS = 0) are recorded zeros and kept
Consistent	Date format and encoding differ by year (handled); same category sets; 68 common hospitals for year comparisons
Conformed	Codes match FONASA and MINSAL dictionaries (200717 missing from FONASA's sheet)
Current	Latest discharge Dec 2024; fit for annual review, not daily bed management
Comprehensive	GRD hospitals only; no bed-waiting, causes of delay, readmissions or costs

Closing the benchmark definition

Network reference per IR-GRD code, pooled 2023-2024

1 Stratum

IR-GRD code; < 30 stays → base GRD (28 codes); still < 30 → excluded (5)

2 Cut-off

Q3 + 1.5·IQR of stay in the stratum; above = extreme (outlier)

3 Expected stay

Mean stay of the stratum's ordinary (inlier) cases

4 IEMA

Σ observed ÷ Σ expected over a hospital's inliers, with 95% interval

Limits

- Relative, not absolute: if the whole network stays too long, IEMA is still 1. The ministerial norm is not public
- GRD and severity miss social and waiting factors: a high IEMA is a signal for review, not proof
- Intervals assume independent episodes, so they are optimistic; hospitals near 1 are not ranked

Does it survive a changed assumption?

Six variants, one change each; hospital ranking 2024

Variant	Rank corr.	Class changes	% outlier
V0 Main definition	1.000	0	6.3%
V1 Cut-off Q3 + 3.0·IQR (only far-out stays are outliers)	0.986	7	2.5%
V2 Year-specific benchmark (2023 and 2024 separately)	0.999	6	6.5%
V3 Exclude deaths and transfers out (reference and evaluation)	0.978	5	6.2%
V4 Peer-group benchmark by hospital complexity level	0.979	4	6.3%
V5 Exclude national institutes from the reference population	1.000	0	6.2%
V6 Minimum stratum size 100 instead of 30	1.000	0	6.3%

≥ 0.98

rank correlation with the main definition in every variant

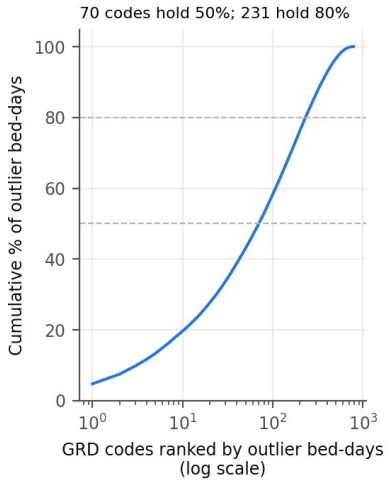
≤ 7

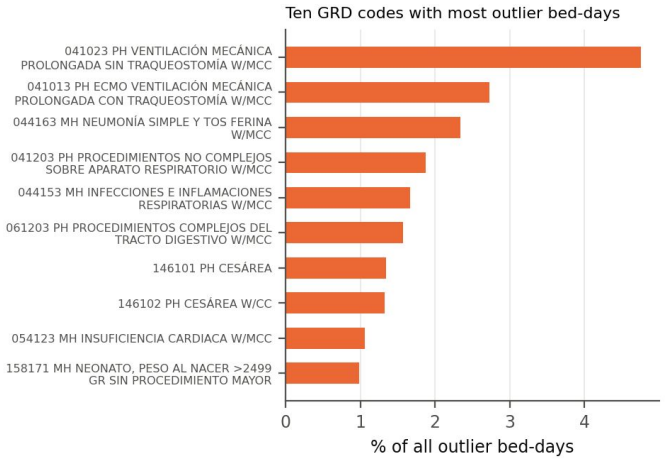
of 72 hospitals change class; all within 0.09 of IEMA = 1

The cut-off rule changes the level of the outlier KPI, not the ordering: always show the rule with the KPI.

Extreme stays: few stays, a quarter of bed-days

6.3% of stays, 25.4% of bed-days in 2024 (mean 28.5 vs 5.6 days)



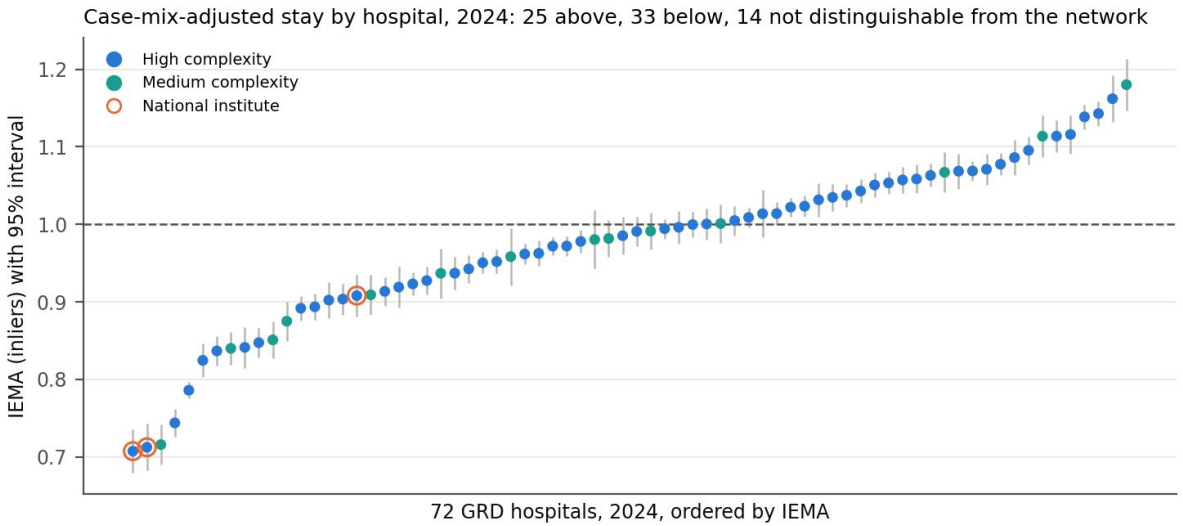


What we observe

- 70 of 764 GRD codes hold half of outlier bed-days; 231 hold 80%
- Top: prolonged ventilation, pneumonia with major complications, complex digestive surgery, caesarean section
- Emergency admissions exceed their cut-off more than expected (O/E 1.19); elective far less (0.45)
- Stays ending in transfer out or other (11.1%) and home hospitalisation (10.4%) are the most often extreme: an association, not a cause

Hospitals: adjusted stay with intervals

IEMA 2024, 72 GRD hospitals; 1 = network behaviour for the same case mix



25

hospitals above the network (interval > 1)

33

below the network (interval < 1)

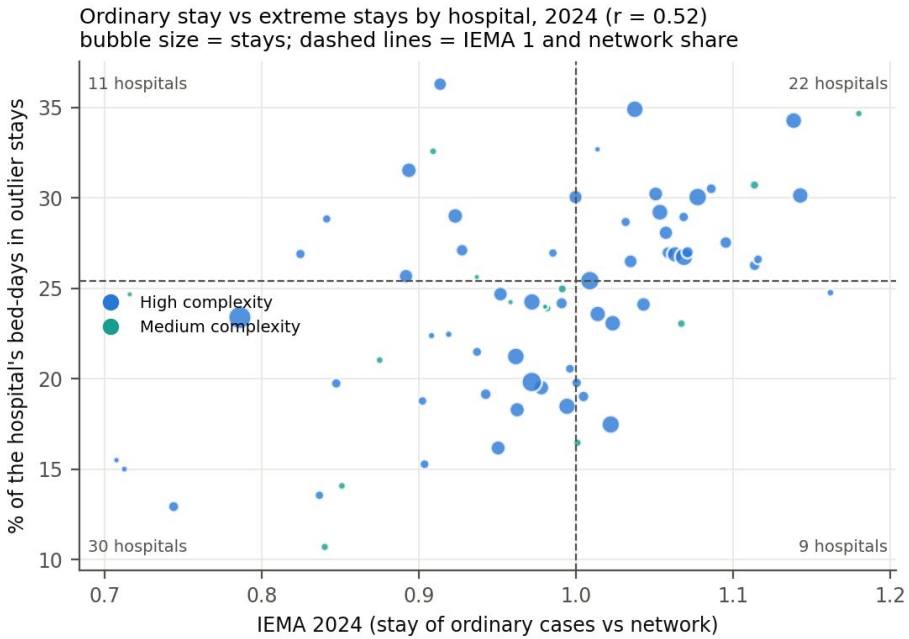
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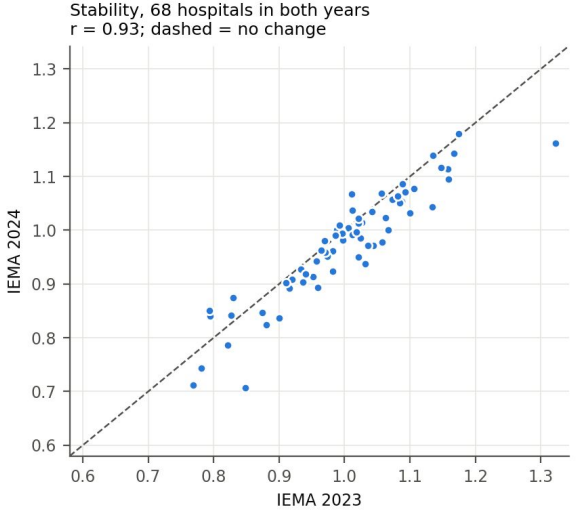
not distinguishable: not ranked

Range 0.71-1.18. Crude stay tracks case mix (r = 0.46); IEMA does not (r = -0.08). The 3 national institutes are below the network (0.71, 0.71, 0.91), the opposite of our W1 hypothesis.

Ordinary vs extreme stays, and stability

Ordinary stay and extreme stays are related but distinct (r = 0.52)





IEMA 2023 vs 2024: r = 0.93; 58 of 68 hospitals keep their class. The two signals may call for different reviews: discharge processes for ordinary cases, case review for prolonged stays.

Conclusions within the evidence

Observed

25 hospitals keep ordinary cases longer than the network and 33 shorter; stable and robust. Half of extreme-stay bed-days fall in 70 of 764 GRD codes.

Interpretation

Compatible with differences in discharge processes, post-acute access or case mix not captured by GRD. The data cannot choose.

Cannot establish

Efficiency against a clinical standard, avoidable days, quality, causes or effects of any intervention.

Implication

Start reviews in hospitals clearly above 1 and their GRDs with most days beyond the cut-off. Next: official norm, delay causes, later 2024 extract.

Continuity: ML task and dashboard

Proposal only: no model trained, no algorithm chosen, no performance claimed

ML: flag extreme-stay risk at admission

- Unit: inpatient episode at admission
- Label: stay above its GRD cut-off (≈ 6.3%, stable)
- Inputs: age, sex, admission type, origin, admitting unit, insurance, hospital, month
- Excluded (leakage): GRD, severity, diagnosis, discharge fields
- To review: field timing; label when GRD unknown at admission; imbalance
- Intended use: start discharge planning on day one

Dashboard for bed management

- Users: FONASA GRD team; hospital and Health Service bed managers
- KPIs: IEMA with interval and class; % bed-days in extreme stays; outliers O/E; days beyond cut-off
- Views: intervals, quadrant, GRD list, monthly trend, definition panel
- Filters: year, month, hospital, complexity, GRD, admission and discharge type
- Rules from EDA: no ranking near 1; rule shown next to outlier KPI